

AI-Powered Telecom Customer Churn Prediction

The background is a light blue gradient. On the left, there is a large, semi-transparent blue sphere. In the bottom right corner, there is a blue curved shape that resembles a stylized 'C' or a partial circle, with a small, dark blue sphere positioned inside it.

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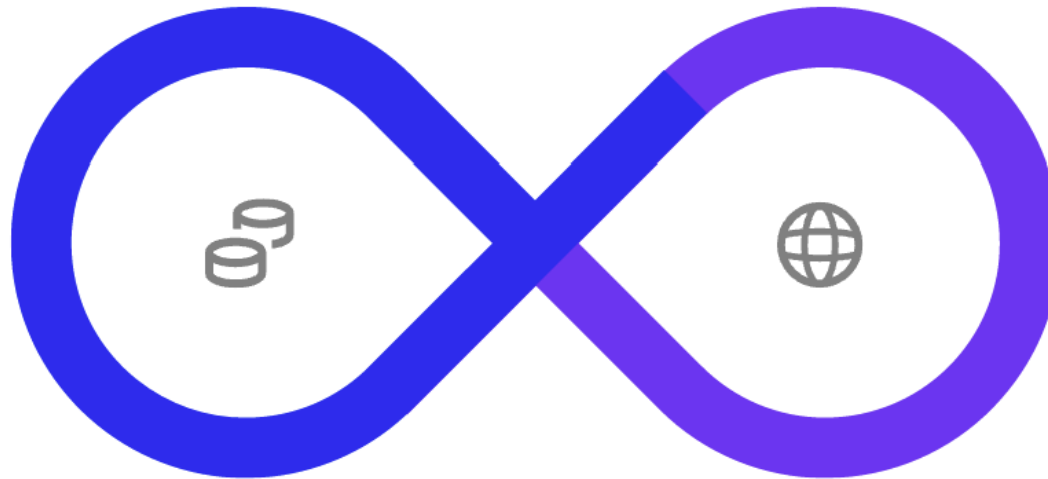
From Problem to Predictive Solution

Business Problem & Objectives

Churn Challenge & Business Impact

Telecom faces high customer churn, raising revenue volatility and pressure on margins.

Retention is cheaper than acquisition, directly affecting recurring revenue stability and long-term profitability.



Predictive Goal & Strategic Value

Primary objective is early churn prediction to trigger targeted, proactive retention interventions. This positions churn analytics as a strategic business intelligence initiative guiding data-driven commercial decisions.

Dataset & Data Preparation Overview

Dataset structure & churn label definition

Dataset includes one hundred thousand customers and ninety-eight features covering usage, revenue, device, and demographic data, with a clearly defined binary churn label target.

Cleaning, engineering, and EDA data pipeline

Pipeline addresses missing values systematically, emphasizes targeted feature engineering, follows structured exploratory data analysis steps, and directly links overall data quality to downstream model reliability.



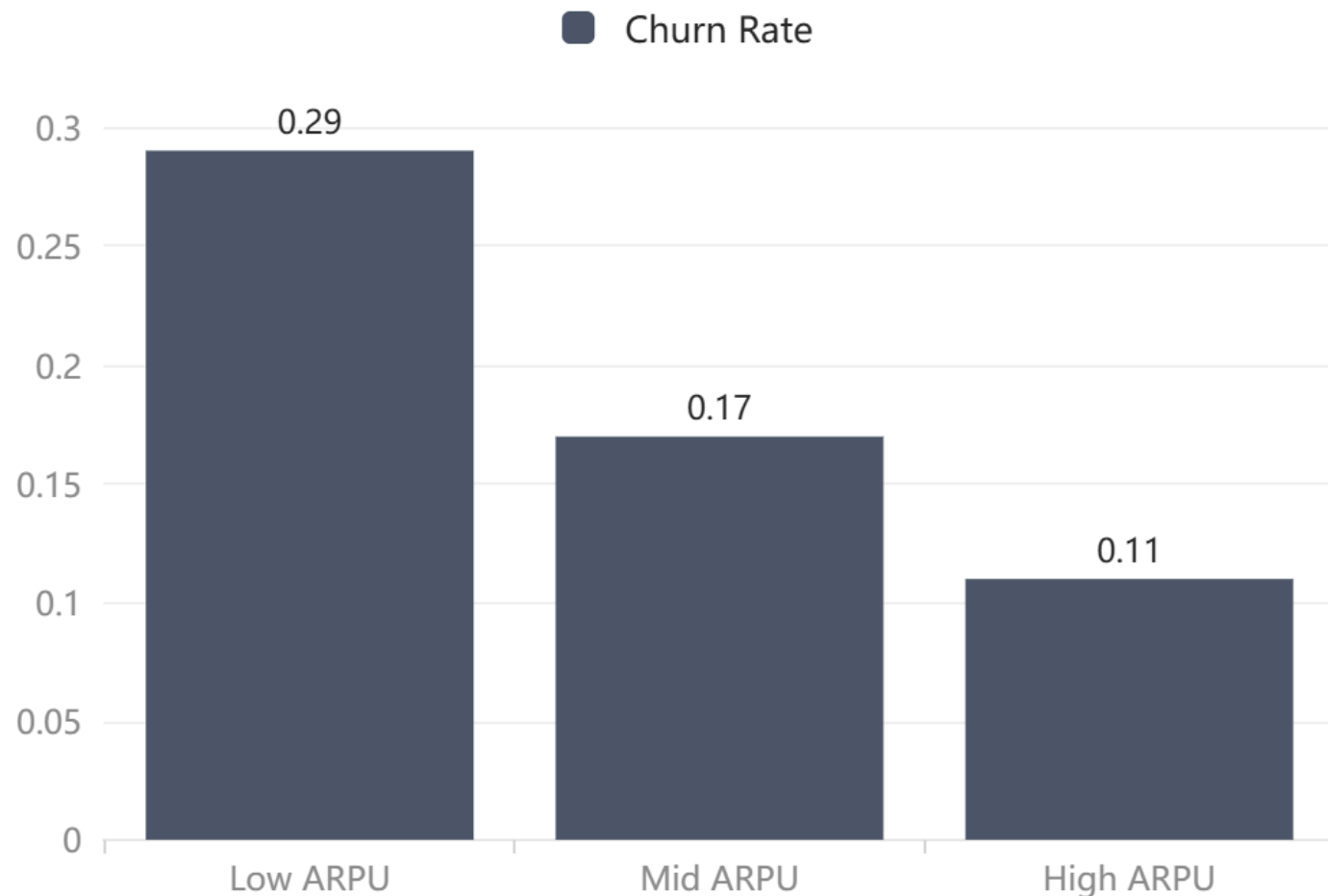
Key Customer Behavior Insights

Churn Drivers and Usage Patterns

Analysis shows churn risk increases with device age, particularly beyond mid-life cycles. Low-usage customers churn disproportionately, suggesting engagement intensity is a critical early-warning behavioral signal.

Revenue Signals and Risk Segmentation

Declining monthly revenue strongly correlates with churn, especially in already low-usage segments. Heatmaps and churn-rate graphs reveal distinct high-risk clusters, guiding feature engineering and targeted retention segmentation.



Machine Learning Solution Design

Feature Engineering for Business Value



Device and Engagement Risk Indicators

Define device risk score from failure patterns and complaints, then derive engagement score using usage frequency, service adoption, and support interactions to quantify customer stickiness and dissatisfaction risk.



Churn Flags, Segments and Business Alignment

Describe high-risk flag thresholds, segment customers into risk tiers, map network and billing signals into interpretable indicators, ensuring features directly support marketing retention plays and service interventions.

Model Evaluation & Final Selection



Comparative model assessment

Evaluated Random Forest, LightGBM, and CatBoost using AUC and interpretability, comparing consistency, overfitting risk, and operational feasibility across historical churn datasets and simulated deployment conditions.



CatBoost as final model choice

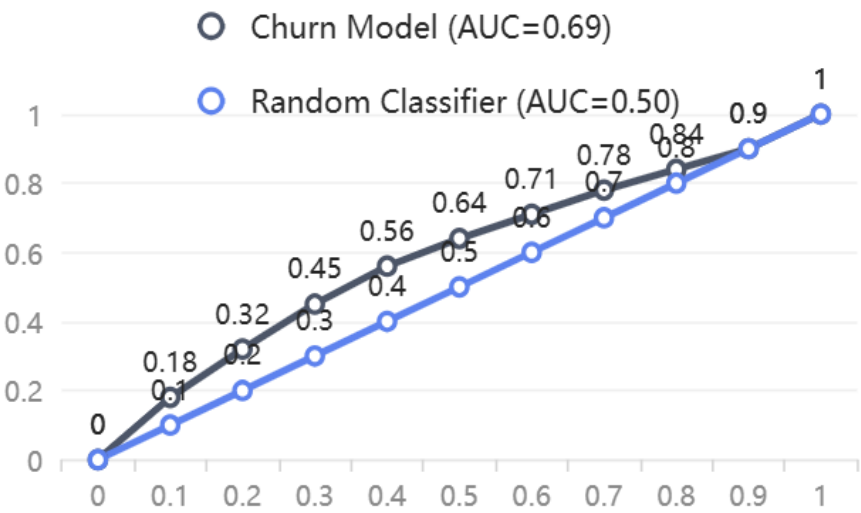
Selected CatBoost for superior categorical handling, higher AUC, more stable predictions, and seamless integration with the existing feature-engineering pipeline for scalable, production-grade churn prediction.

Model Performance

Churn Detection

Average Value	YoY Growth
500	10%

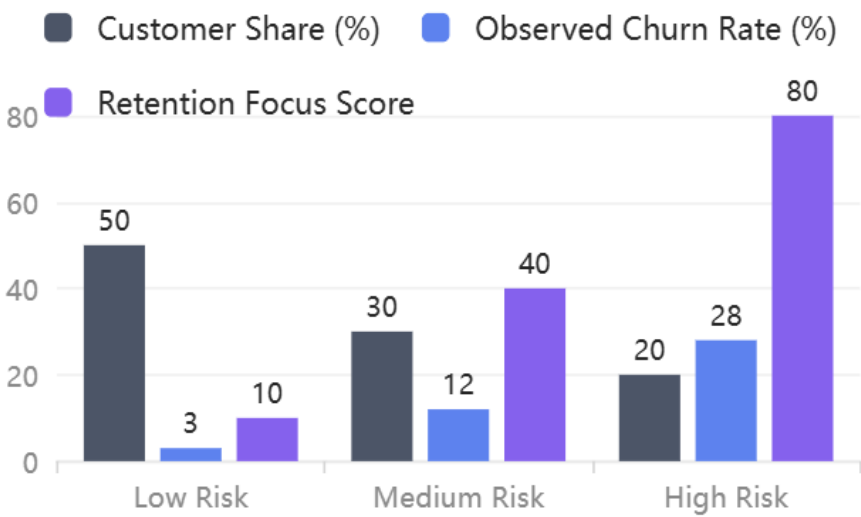
The model achieves an AUC of 0.69, indicating solid lift over random guessing and practical value for early churn detection. By ranking customers by churn probability, teams can spot high risk segments sooner and act before customers leave.



Business Impact

Average Value	YoY Growth
500	10%

The ROC curve illustrates the trade off between true and false positive rates, guiding threshold choices by risk group. Clear segmentation of low, medium, and high risk customers improves retention prioritization and links model accuracy to measurable revenue protection.



From Analytics to Business Impact

Churn-Based Customer Segmentation

High-Risk and Medium-Risk Customers

High-risk: older devices, declining usage, noticeable revenue drops. Medium-risk: moderate device age, steady but flattening usage, stable short-term revenue yet emerging behavioral churn signals.

Low-Risk Segment and Business Impact

Low-risk: newer devices, high engagement, growing revenue. Use a churn risk matrix combining device, usage, revenue to guide targeted retention, marketing campaigns, and service operations planning.

Targeted Retention Strategies



Personalized Campaign Design

Define device upgrade campaigns for legacy users, tailored retention offers, and special plans aligned to customer profiles and value, ensuring relevant, timely outreach across digital and assisted channels.

Present loyalty incentives for at-risk high-value customers, plus usage-boost programs including bonus data and promotional bundles, linked to proactive engagement triggers from behavioral and network usage insights.

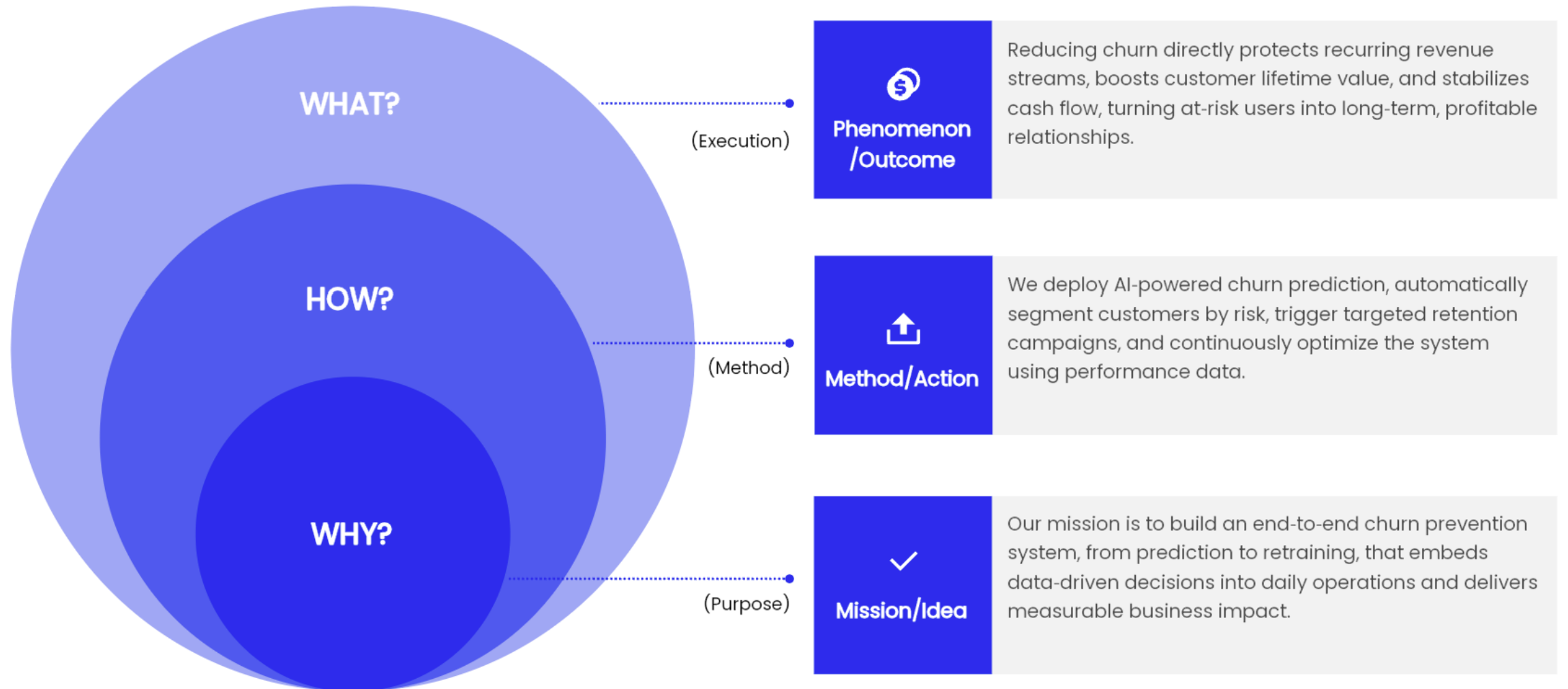


Risk-Based Execution & Measurement

Map every retention action to specific churn risk segments and probability bands, using AI-driven scores to prioritize outreach, optimize resource allocation, and customize journey paths by segment.

Emphasize closed-loop campaign measurement, tracking uplift, churn reduction, and incremental revenue, feeding results back into models to refine risk scores, offers, and targeting precision continuously.

Churn impact

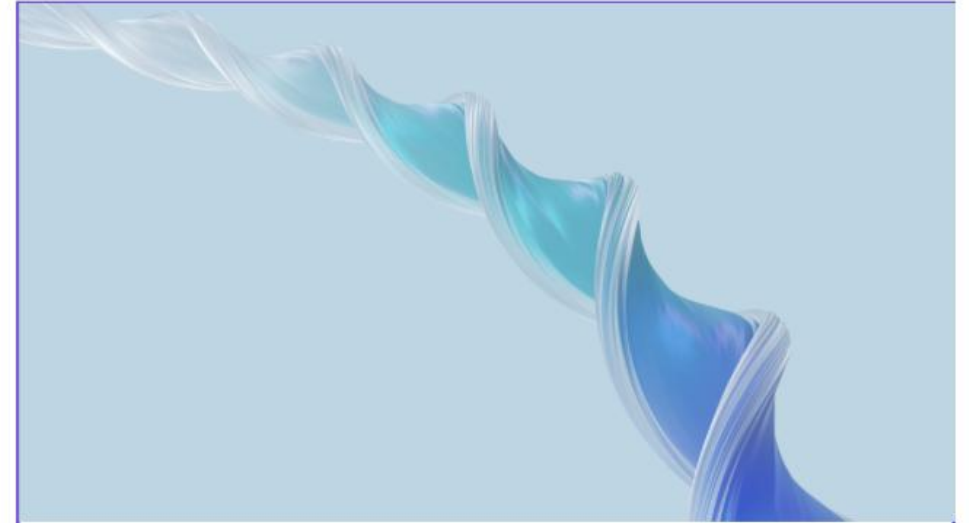


Conclusion & Future Roadmap



Achievements & Business Impact

Deployed churn model identifying high-risk customers, driven by business-centric features. Enabled targeted retention actions that protect revenue, enhance loyalty, and strengthen decision-ready insights.



Future Enhancements & Vision

Plan Optuna-based tuning, real-time churn dashboards, and automated retention engines for always-on interventions, advancing analytics maturity toward a fully proactive, impact-focused BI ecosystem.